

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Use the classical electron energyâĢmomentum relation for relativistic de Broglie wavelength :
 $\hat{p} = h/(\hat{\lambda}(2mE)^{1/2})$,
where h is PlanckâĢs constant (in ergÂĢs), m is the electron mass (9.11×10^{-28} g), c is the
speed of light (3.00×10^{10} cm/s), and $E = 1 \text{ eV} = 1.60 \times 10^{-12}$ erg .

Calculate the factor :
 $\hat{\lambda}(2mE/c^2)^{1/2} = \hat{\lambda}[(2 \times 9.11 \times 10^{-28} \times 1.60 \times 10^{-12})/(3.00 \times 10^{10})^2]^{1/2} = 1.875 \times 10^{-23} \text{ gÂĢcm/s} .$

Then $\hat{p} = h/(\hat{\lambda}(2mE)^{1/2}) = (6.625 \times 10^{-27} \text{ ergÂĢs})/(1.875 \times 10^{-23} \text{ gÂĢcm/s} \times 3.00 \times 10^{10} \text{ cm/s}) = 1.23 \times 10^{-7} \text{ cm} .$

Correct answer: G. 1.23×10^{-7} cm
</think>
<answer>
G. 1.23×10^{-7} cm
</answer>